

AUDIT REPORT 2 OF 3 · ROUND 4

UX Report — Tick & Taka (round 4)

Android app for one person's day and money · re-audit after 36 fix commits

Project	Tick & Taka (monorepo: apps/mobile , apps/api , packages/shared)
Date	2026-10-07
Commit	801dd5f (branch fix/audit-2026-10-06)
Previous audits	Round 1 at 3b5438b : 64/100. Round 2 at df29d67 : 83/100. Round 3 at 7e2bfeb : 89/100, 11 open findings (docs/audit/2026-10-07-r3/UX_Report.html).
Auditor role	UX Lead
Yardstick	PRODUCT.md (user, five principles, accessibility commitments) and DESIGN.md (token table, type, layout, components), then general best practice
Mode	Read-only static review of the code. The app was not run: the only emulator image has no Google Play services and sign-in is Google-only. The screenshots in docs/screenshots were last changed in 1f25750 (2026-10-05) and predate every fix in rounds 2 to 4.

1. Executive summary

92/100

Round 3: 89. Round 2: 83. Round 1: 64.
Round-3 findings: **10 Fixed**, 0 Partially fixed, **1 Not fixed** (UX-029, screenshots), 0 Regressed.
Open now: 0 Critical, 0 High, 1 Medium, 9 Low (10 in total).

Every round-3 finding that touched the product was fixed in code, most with a test that will catch a relapse. The Undo after "Paid", "Received" or "Skip" now goes through the saved queue and holds across a restart. The widget shows one-tap quick-logs at its default size, from a tested layout function. `Chip` must declare its role, on/off settings are switches, the tab bar is a tablist, calendar days are 49 dp, chart labels are 13 px, and pickers show values set elsewhere as themselves ("9:00 pm"). `DESIGN.md` now agrees with itself and records the new trade-offs.

One Medium is left: the JSON export that the Delete-account page offers as step 1 is deleted as soon as the share sheet returns, and nothing confirms a copy was saved (device check needed). The nine Lows are smaller: bare " + - " glyphs on the default-size widget, mint on a few buttons and switches, faint Plan-calendar dots, and a few copy and form edges. The README screenshots are still undated.

Severity	Critical	High	Medium	Low	Total
Open findings (round 4)	0	0	1	9	10
Round 3, for comparison	0	0	2	9	11

Top 5 to fix first

Severity	Issue
Medium	UX-052 The export is deleted the moment the share sheet returns, and nothing says it was saved; it is step 1 before account deletion (<code>features/settings/use-export-data.ts:90-112</code>). Unverified
Low	UX-049 At 4x2 the widget shows Task and Expense as bare " + " " - " (a new user sees " + - ►🔔"); the fit check counts padding the pills don't have (<code>features/widget/widget-layout.ts:27, 66-69</code>).
Low	UX-050 A <code>money</code> button fills "Paid" (a bill) and "Log as one expense" in money-in green; three switches use a mint track at 2.21:1 (<code>components/ui/button.tsx:17</code>).
Low	UX-051 Plan-calendar task and bill dots are the only visual cue, in bright tokens: coral 2.55:1, 2.19:1 on today (<code>features/plan/calendar-view.tsx:130-145</code>).
Low	UX-054 A write the proxy refuses as too large is dropped, but the snackbar says "Saved on this phone..." (<code>lib/error-copy.ts:62</code> , <code>lib/outbox-failure.ts:13</code>).

Why 92 (built in section 2): no Critical or High and one Medium put it in the 90s. It is not higher because that Medium sits on the account-deletion path, several round-4 gains (widget at real sizes, TalkBack roles, share targets) are device-only, and the colour rule still has gaps outside grape and mango.

2. Scope and method

What was reviewed

- **Intent:** `PRODUCT.md` (unchanged since round 1) and `DESIGN.md`. The `DESIGN.md` changes since round 3 (`8d9b7e4`, plus lines added by `5fb18d3` and `74d944e`) were each checked against the code and judged on whether they are explicit, reasoned decisions (section 4.1).
- **Round 3:** all 11 open findings in `docs/audit/2026-10-07-r3/UX_Report.html`. Each was checked against the code at `801dd5f`, not against commit messages (36 commits in `git log 7e2bfeb..HEAD`, 89 files changed under `apps/mobile/src`).
- **Code read in full:** `theme/palette.ts`, `global.css`, `tailwind.config.js`; `components/ui/` `chip`, `a11y-actions`, `segmented`, `picker-field`, `picker-options`, `date-field`, `toggle-row`, `icon-button`, `month-stepper`, `chart-axis`, `calendar-grid`, `card`, `button`, `snackbar`, `sync-status`; `components/navigation/tab-bar.tsx`; all of `features/widget/`; `features/today/next-up.tsx`; `features/money/` `recurring-pay`, `recurring-sheet`, `account-sheet`, `goal-sheet`, `debt-sheet` (switch), `budget-edit-sheet` (switch), `money-calendar`, `events-screen`; `features/plan/calendar-view.tsx`, `week-planner`; `features/tasks/task-sheet.tsx`, `task-more-fields`; `features/time/` `time-entry-sheet`, `time-tracking`, `focus-stats`; `features/review/` `insights`, `ai-usage`; `features/settings/` `delete-account`, `delete-account-rules`, `use-export-data`, `data-section`, `account-section`, `setting-row`, `planning-` and `notification-settings`; `features/habits/habit-sheet.tsx`; `features/quick-add/quick-add-sheet.tsx`; `lib/` `outbox`, `outbox-queue` (cancel, discard, restore), `outbox-failure`, `error-copy`, `api`, `http`, `notify`, `snack-timing`, `use-today`, `user-time`, `export-file`, `user-data`, `receipts`.
- **Tests read for intent:** `widget-layout`, `colour-meaning`, `type-scale`, `calendar-grid`, `picker-options`, `recurring-pay`, `time-zone-use`, `contrast` (outline), `outbox-queue` (Discard/Undo).

Scripted checks (scratch in `_ux_tmp/`, deleted afterwards)

- **Contrast:** `python3 -I contrast.py` parsed `palette.ts` (base and accent themes), `global.css` and the `DESIGN.md` colour table, rebuilt the widget palette as `widget-colors.ts` does, and computed WCAG 2.x ratios for 254 pairs (Appendix A). Token diff: 0 mismatches in 70 comparisons (17 CSS variables × 2 schemes against the palette, 18 `DESIGN.md` rows × 2 schemes against the palette).
- **Widget layouts:** `bun run layouts.ts` imported the real `widgetLayout()` and printed the pill row for 9 sizes × 4 states (section 7.2, UX-049). This runs the shipped function. It does not render the widget.
- **Accessibility scan:** `python3 -I scan.py` over `apps/mobile/src`: 60 `<Pressable>`/`<SwipeRowPressable>` tags in 45 files, checked for role, label and size, plus counts of accessibility props, field components and text sizes (Appendix B).
- Reused, not re-run (from `_raw/`): Biome (555 files, exit 0), typecheck (3 workspaces, exit 0), tests (shared 247, mobile 291, api 236 pass, 0 fail).

What could not be verified

There was no device or emulator run, so all of the following are Unverified:

- How the widget actually renders at real launcher cell sizes (Pixel, Samsung), including how its label-width estimate compares with the real font and with large system font sizes.
- What TalkBack reads for the new radio/checkbox chips, `ToggleRow` (label plus switch), the tablist that holds a plain Quick-add button, and the snackbar's Undo and Dismiss.

- Whether share targets (Drive, Gmail, Files) can still read the export after the app deletes it (UX-052).
- The size of the RN `Switch` hit area on Android (UX-053).
- Layout at 200% font size, including the 13 px chart labels, which scale with the font.
- Reanimated springs under the system reduce-motion setting (unchanged: 6 files use `useReducedMotion`, and Reanimated 4 defaults to `ReduceMotion.System`).

Screenshots: only `login.png` is embedded (section 4), because `app/login.tsx` hasn't changed since round 3 and still matches it. Every other screenshot shows pre-fix layouts. The widget is shown as a schematic built from `widgetLayout()` output, not as a screenshot.

Score breakdown

Area	Weight	R3	R4	Points	Main reason for the change
Fit to the 5 principles + accessibility commitments (avg 4.5/5)	25	87%	90%	22.5	Thumb-size gaps closed (calendar 49 dp). Colour residue moved from grape/mango to mint (UX-050, UX-051).
Core task flows (taps, friction, undo)	20	93%	95%	19.0	Pay Undo survives a restart, quick-log at default size, Discard has Undo. The export hand-off is weak (UX-052).
Nielsen heuristics (avg 4.7/5)	15	90%	94%	14.1	Recognition (off-list values) and minimalism (chip walls gone) up; widget glyphs and the README hold back two rows.
Visual design and token consistency	10	85%	90%	9.0	0 token mismatches, DESIGN.md consistent, one vocabulary for fields. Undocumented <code>money</code> button and mint switches.
Accessibility (contrast, SR, scaling, motion)	15	88%	91%	13.65	Roles fixed and enforced by the type system; 13 px floor tested. Plan-calendar dots and mint tracks below 3:1; TalkBack Unverified .
Mobile ergonomics (targets, reach, keyboard)	5	88%	93%	4.65	Calendar cells 49 dp without overlapping slop; IconButton 48 dp. Switch rows only toggle on the switch (UX-053).
Content, microcopy, feedback and delight	10	95%	93%	9.3	12-hour times, off-list values in words. Re-judged: export gives no success feedback (UX-052), and the 413 copy is wrong (UX-054).
Total				92.2 → 92	No Critical or High, so no deduction

Confidence. The score rests on the code and on the tests that pin it. If a device run shows the widget's estimate clipping labels, or share targets losing the export, UX-049 and UX-052 go up and the score drops by 1 to 3 points. If Drive and Gmail read the export before the app deletes it, UX-052 becomes Low (a missing confirmation) and the score rises by about 1.

3. Previous findings status (round 3 → round 4)

Each of the 11 open round-3 findings was checked against the code at 801dd5f. "Fixed" means the problem described in round 3 is gone from the code. Device behaviour is still **Unverified** where noted. Paths are relative to `apps/mobile/src/` unless they start at the repo root.

ID · R3 sev.	Round-3 finding	Status	Evidence now
UX-042 Medium	Offline "Paid" Undo held only in memory	Fixed	<code>features/money/recurring-pay.ts:36-52</code> : the pay goes into the saved queue, and Undo first tries <code>send.cancel(r => r === pay)</code> . If the pay hasn't gone out, it is removed from the queue (<code>lib/outbox-queue.ts:283-294</code> , which also searches set-aside writes and persists). Otherwise <code>POST /recurring/:id/unpay</code> is queued behind it. The unpay needs only the phone-made <code>transactionId</code> and <code>dueAt</code> , not the reply (<code>apps/api/src/modules/recurring/routes.ts:33</code> , <code>service.ts:193</code>). Used by Today (<code>today/next-up.tsx:49-62</code>) and the bill sheet (<code>money/recurring-sheet.tsx:115-138</code>). <code>test/recurring-pay.test.ts:39-110</code> covers offline cancel, "may have gone out" and after landing. "Took back Rent" is now true when it is said.
UX-043 Medium	Widget quick-logs hidden at default size; labels clip at 250 dp	Fixed	The new pure <code>features/widget/widget-layout.ts</code> puts up to two quick-logs in the single row beside Task and Expense at "normal" height (160-239 dp; : 148-164), and gives them a row of their own at ≥ 240 dp. Pills never go below 48 dp. A label that doesn't fit becomes a glyph, a quick-log that doesn't fit is left out, and text has <code>truncate="END"</code> (<code>safe-to-spend-widget.tsx:80-85</code>). 9 tests (<code>test/widget-layout.test.ts</code>) and DESIGN.md:79 record the trade. Side effect: UX-049. Rendering on real launchers Unverified .
UX-036 Low	Single-choice chips announced as buttons; tabs without tablist	Fixed	<code>components/ui/chip.tsx:18-27</code> : <code>choice</code> is now required, and an <code>action</code> chip can't take <code>selected</code> (a type-level union). All 15 call sites declare it: 11 single, 1 multi, 3 action. Every single-choice row sits in a <code>radiogroup</code> (13 <code>radiogroup</code> declarations in the code). The on/off chips (Billable ×2, top three, Urgent) became <code>ToggleRow</code> switches (<code>components/ui/toggle-row.tsx</code>). The tab bar's capsule is <code>accessibilityRole="tablist"</code> (<code>tab-bar.tsx:106</code>). Quick-add sits inside it as a plain button, which is acceptable. TalkBack reading Unverified .
UX-013 Low	Chip rows remained in five sheets	Fixed	Account type and currency, bill currency and "Remind me", goal/event/category/area emoji, and time-entry "How long" are now <code>PickerField</code> s (list or emoji grid). "Next due", "Deadline", "Started", "From/To" and the debt reminder are <code>DateField</code> s. "Billable" is a switch. The chip rows left are short single-choice rows the DESIGN.md chip rule allows (project status, shopping list names, the account to pay from). 47 <code>PickerField</code> , 9 <code>DateField</code> and 15 <code>ToggleRow</code> uses. Commit 35df896 matches the code.

ID · R3 sev.	Round-3 finding	Status	Evidence now
UX-044 Low	Colour drift outside the colour test	Fixed	AI cost bars are <code>colors.muted</code> and the progress bar is ink, or coral when over the cap (<code>review/ai-usage-screen.tsx:77, 116</code>). Keypad operators are <code>bg-line</code> (<code>components/ui/amount-keypad.tsx:73</code>). Thinking dots are muted (<code>assistant/thinking-indicator.tsx:74</code>). Switch "on" is ink (<code>toggle-row.tsx:33</code>). <code>test/colour-meaning.test.ts:66-106</code> now matches classes, <code>colors.x</code> , tone strings and ternaries, with allow-lists for grape and mango, and checks the regex itself. Gaps remain: the test covers only grape and mango, so mint misuse got through (UX-050).
UX-045 Low	Pickers say "Choose" for values set elsewhere; 24-hour times	Fixed	<code>components/ui/picker-options.ts:13-21</code> with <code>currentOption</code> lists an off-list value first, labelled by <code>describe</code> . It is used by <code>PickerField</code> (<code>picker-field.tsx:66</code>) and passed for daily goal, free time, week start, work/break, remind days and every time. <code>clockOptions</code> / <code>clockLabel</code> give "9:00 pm" (notification settings :25-44, habit "Nudge me at"). 6 tests. Residue: the habit "How often" picker has no <code>describe</code> (UX-057).
UX-046 Low	Discard on a stuck change: one tap, no Undo	Fixed	<code>components/ui/sync-status.tsx:61-80</code> : Retry and Discard are 24 dp apart (<code>gap-6</code>). Discard drops the whole set-aside group and shows <code>Discarded "..."</code> · Undo, which calls <code>outbox.restoreStuck(group)</code> (<code>lib/outbox-queue.ts:138-160</code>). Tested (<code>outbox-queue.test.ts:429</code>). The copy is a little odd: UX-054.
UX-047 Low	Chart axis labels at 10 px	Fixed	One <code>useChartAxis()</code> (<code>components/ui/chart-axis.ts:4-23</code>) sets 13 px Nunito SemiBold muted labels for all three charts (insights, focus stats, AI usage). <code>test/type-scale.test.ts</code> fails on any <code>fontSize</code> or <code>text-[Npx]</code> below 13 except the 12 px tab label. My scan agrees: the only size below 13 is <code>tab-bar.tsx:77</code> .
UX-048 Low	DESIGN.md contradicts itself	Fixed	DESIGN.md:21 now reads "Primary action, FAB, Tiki; never a selected state". :74 says "secondary = surface with a line-strong border" and names the <code>time</code> variant. Also recorded: dates stay 0-9 (:47), month-grid bleed (:68), chip roles (:75), switches (:76), off-list picker values and <code>DateField</code> (:77), and widget size classes (:79). Judged in section 4.1.
UX-040 Low	Calendar cells ~43-46 dp; overlapping hitSlop	Fixed	<code>components/ui/calendar-grid.ts:10-16</code> : <code>marginHorizontal: -12</code> and no side padding, so $(360 - 40 + 24) / 7 = 49.1$ dp per day. Used by both calendars (<code>money-calendar.tsx:74</code> , <code>plan/calendar-view.tsx:92</code>). The <code>hitSlop</code> is gone (neither file has one). <code>test/calendar-grid.test.ts</code> pins ≥ 48 dp and keeps the grid clear of the screen edge.
UX-029 Low	README screenshots predate the fixes	Not fixed	<code>docs/screenshots/*.png</code> were last changed in 1f25750 (2026-10-05). <code>README.md:10-14, 33-72</code> still embeds them with no date or note. The README's only changes this round are the assistant-Undo and audit-allowlist paragraphs. Carried as UX-029.

Tally: 10 Fixed · 0 Partially fixed · 1 Not fixed (UX-029) · 0 Regressed.

Commit-message accuracy. I spot-checked the claims behind each fix (35df896 , 7995f56 , a35e73b , cd4f544 , 74d944e , 5fb18d3 , 9d3e708 , 6010dc9 , 6581c93 , 8d9b7e4 , 1fc2209), and each matches the code. Unlike round 3, no commit claims more than it did.

4. Fit to product principles

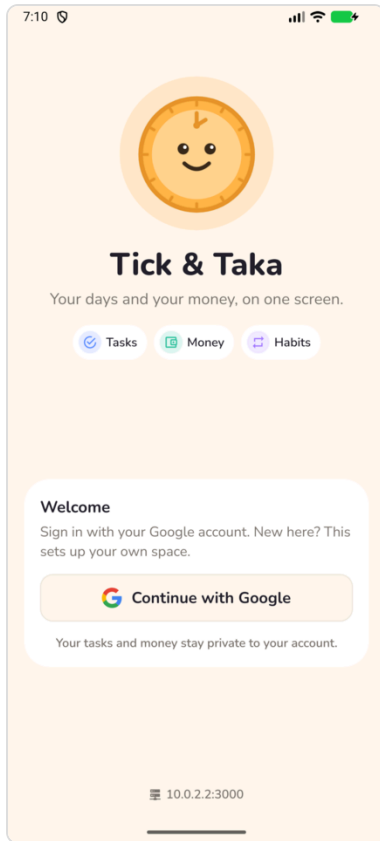
Principle (PRODUCT.md)	R3	R4	Evidence
1. Today first, one glance	5	5	Unchanged and right: the header with "Next: ...", then safe to spend, top three, next up, habits and later (<code>app/(tabs)/index.tsx</code>). "Today" is now worked out in the user's zone everywhere (<code>lib/use-today.ts</code> , tested by <code>time-zone-use.test.ts</code>). Bills say "Was due Mon 5 Oct".
2. Capture beats organise	4	4	Up: one-tap quick-logs now appear on the widget at its default size. Not yet met: quick-add still shows the Kind, Account and Category pickers on every money entry instead of "only when the parser needs help" (<code>quick-add/quick-add-sheet.tsx:159-200</code>). The widget's glyph-only actions cost a little recognition (UX-049).
3. Big, calm, thumb-sized	4	5	No pressable below 48 dp without slop (Appendix B). Calendar days 49 dp, IconButton 48 dp, PickerField and DateField 56 dp, widget pills 48 dp. Left: switch rows only toggle on the switch itself (UX-053).
4. Colour means something	4	4	Grape is limited to habit code and mango to the primary action, FAB and Tiki, both enforced by a widened test. New residue in mint: a <code>money</code> button fills "Paid" (bill) and "Log as one expense" green, and three money-sheet switches have a mint track (UX-050). The plan calendar's coral bill dot is a bright mark used as the only cue (UX-051).
5. Reward, never guilt	5	5	No "Overdue", no zeros, and stuck changes are described calmly. Discard now offers Undo instead of an alarm.
Accessibility & Inclusion commitments	4	4	Every text pair passes AA (146 pairs, lowest 4.92 (dark: skyText on sky/15 over card)). Roles are enforced by types, and the 13 px floor and 48 dp calendar are enforced by tests. Not met: 5 non-text pairs below 3:1 (plan-calendar dots, mint switch tracks; Appendix A). TalkBack behaviour Unverified .

Average of the six rows: **4.5 / 5** (round 3: 4.33).

4.1 Decisions recorded in DESIGN.md since round 3: explicit, or a moved yardstick?

Decision	Where	Verdict	Why
Mango is "never a selected state"	:21, :34	Accepted	This tightens the rule. The code agrees: chips, picker grid and switches select with ink, and keypad operators are <code>bg-line</code> . Small gap: the test's mango allow-list also covers the pull-to-refresh spinner and the navigation theme's primary (<code>colour-meaning.test.ts:57-58</code>), which the role line doesn't name. Add "spinners" to :21 or make the spinner muted.
Buttons: secondary = line-strong outline; a <code>time</code> variant	:74	Accepted, incomplete	Matches <code>button.tsx:16-18</code> . But the code has a fifth variant, <code>money</code> (mint fill), that the list doesn't mention and that conflicts with :24 and :34. Raised as UX-050.
Chips are actions, radios or checkboxes, never on/off	:75	Accepted	Enforced by the <code>ChipProps</code> union type, so it can't drift silently.

Decision	Where	Verdict	Why
Switches: on = ink track, off = line-strong	:76	Accepted, not fully applied	<code>ToggleRow</code> follows it. Three hand-rolled <code>Switch</code> es in money sheets don't (UX-050).
Off-list picker values listed as themselves; <code>DateField</code> for dates	:77	Accepted	Implemented once in <code>picker-options.ts</code> and tested. The task sheet's "Deadline" is still a <code>PickerField</code> with a "Pick a date..." option rather than a <code>DateField</code> . It works and is consistent with "When", so it isn't raised.
Month grids bleed 12 dp, no overlapping <code>hitSlop</code>	:68	Accepted	A measured, tested layout rule (49 dp) that replaces round 2's flawed slop advice. The grid stays 8 dp from the screen edge.
Widget: quick-logs replace Focus and Tiki at 4×2; glyphs when narrow	:79	Accepted as a trade, with a defect	Putting the fastest path (one-tap logging) ahead of Focus and Tiki fits principle 2, and it is written down with its reason. But "a label is only shown when it fits" is implemented with a padding allowance the stretched pills don't get, so labels drop at sizes where they would fit (UX-049).
Amount digits follow Numbers; dates and times stay 0-9	:47	Accepted	This was asked for in round 3, and the in-app caption says the same.
Tab labels 12 px; FAB off-centre beside Tiki	:61, :73	Accepted (unchanged)	Already judged in round 3, and the 12 px exception is now pinned by <code>type-scale.test.ts</code> .



Screens that changed but can't be re-shot

The account, bill, goal, event, category, area and time-entry sheets now use `PickerField`, `DateField` and `ToggleRow` instead of chip rows. The README's `task.png` and `week.png` show the old chip walls and the old calendar width. The widget in `widget.png` shows the pre-round-3 palette and layout. None of them is used as evidence in this report (UX-029).

`docs/screenshots/login.png`, taken 2026-10-05, before the round-2 to round-4 fixes. `app/login.tsx` hasn't changed since round 3, so this still matches the current screen, apart from the "Account deleted" notice and the dev-only API host line.

5. Core task flows

Taps are counted from Today with the app open. "Type" means one text entry. The stated target is "log in under five seconds". Times are Unverified.

Flow	Steps	Taps	Friction, dead ends, undo (round 4)
Log an expense	+ → "cha 20" → Save	2 + type	Meets the target. Unchanged: "Saved: Expense ₮20" with Undo, offline too.
Quick-log from the widget	Tap "Cha ₮20"	1	Now available at the default 4×2 size, with up to two quick-logs beside "+" and "-". Undo and Keep take the row's first places for 10 minutes. Glyph-only actions: UX-049. Rendering Unverified .
Check safe-to-spend	Open app or glance at widget	0	Unchanged; widget hero mint-text ≥ 5.22:1.
Pay a bill from Today	Next up → "Paid"	1	"Rent logged · Undo". Undo cancels the queued pay or queues an <code>unpay</code> ; both survive a restart (UX-042 fixed). Foreign-currency bills open the sheet for the rate.
Skip a bill	Bill → Skip	2	"Skipped this one · Undo", with the same durable Undo. The sheet footer has four buttons in two rows (Skip/Paid, Delete/Save). That is busy, but each button is labelled.
Log time by hand	Plan → Time → Log time → fields → Save	4-6	"How long", "Started" and "Area" are fields, and "Billable" is a switch. An edited 37-minute entry keeps "37 min" as an option. Area can't be cleared once set (UX-057).
Change quiet hours	Review → Settings → Notifications → field → option	4	Options read "9:00 pm". A time set by Tiki shows as itself.
Look at a calendar day	Money → Calendar → day	2	49 dp cells with no overlapping slop. In Plan's calendar the dots that show tasks and bills are faint in light (UX-051).
A change the server keeps refusing	Card → Retry / Discard	1	Retry and Discard are 24 dp apart, and Discard offers Undo (UX-046 fixed). Later edits to the same record are set aside with it and shown as "...", and 2 later changes to it".
Export my data	Review → Settings → Data → Export all data → share target	4 + pick	No message on success. The file is deleted when the share sheet returns (UX-052).
Delete the account	Review → Settings → scroll → Delete account... → (Export) → type DELETE → Delete my account	4-5 + type	Right amount of friction for the only irreversible action. But step 1 ("Keep a copy") never shows a done state, and its file may be gone before the target app reads it (UX-052).
Talk to Tiki from the widget	Tap "👋"	1	When the server's AI is off, this opens a listening chat that then answers "AI isn't set up yet" (UX-055).

6. Heuristic review (Nielsen's 10)

Heuristic	R3	R4	Evidence
1. Visibility of system status	5	5	The sync line now also covers "not saved on this phone yet" and "can't be read right now" (<code>sync-status.tsx:112-114</code>). "Took back Rent" is now true when shown. One gap: a successful export says nothing (UX-052).
2. Match with the real world	4	5	Times read "9:00 pm" everywhere, including pickers. "Was due Mon 5 Oct". The time zone everything follows is invisible (UX-056), but it is right for the stated user.
3. User control and freedom	5	5	Undo for Discard. Pay Undo is durable. Undo for assistant settings changes (<code>18c7b4a</code>).
4. Consistency and standards	4	4	One field vocabulary (PickerField, DateField, ToggleRow, IconButton, MonthStepper) across 70+ uses. Held back by the three mint switches and the undocumented <code>money</code> button (UX-050), and by widget labels that change between sizes and states (UX-049).
5. Error prevention	5	5	Typed DELETE, and the chip type makes misuse a compile error. The goal deadline silently ignores a past date (UX-057).
6. Recognition rather than recall	4	4	Off-list values now show as themselves. Lost ground on the widget: " + - ► 🔍" have to be learned (UX-049).
7. Flexibility and efficiency	5	5	Natural-language capture, widget quick-log at default size, launcher shortcuts, swipe.
8. Aesthetic and minimalist design	4	5	The last chip walls in sheets are gone. Sheets read as calm stacks of fields.
9. Help users recover from errors	5	4	Stuck writes can be retried, discarded and undone. But a write refused as too large reports "Saved on this phone..." after it was dropped (UX-054).
10. Help and documentation	4	4	README now says which assistant changes have Undo. Screenshots still undated (UX-029).

Average **4.7 / 5** (round 3: 4.5).

7. Visual design, ergonomics, accessibility, content

7.1 Tokens and visual design

- **Token table:** `palette.ts`, `global.css` and the DESIGN.md table agree exactly: 0 mismatches in 70 comparisons. `tailwind.config.js` maps every token through the CSS variables, and `on-mango` is the fixed `#23202B` in both. The tokens are unchanged since round 2.
- **Type scale:** the only text below Caption is the 12 px tab label, which DESIGN.md allows and `type-scale.test.ts` pins. There are no `text-xs` classes and no `allowFontScaling={false}`.
- **Colour meaning:** grape and mango are clean and tested. Mint is the new weak spot (UX-050), and bright sky and coral are used as sole-cue dots in Plan's calendar (UX-051). Smaller cases I did not raise: the swipe tray's "Done" action is mint (`tasks/task-row.tsx:44`), and the settings avatar circle is sky (`settings/account-section.tsx:29-30`).
- **Dark mode parity:** unchanged and sound. Every dark text pair passes, and ToggleRow's ink-on track works in both themes (thumb on track 16:1).

7.2 The widget at launcher sizes (computed, not rendered)

`widgetLayout()` run on the shipped code (`_ux_tmp/layouts.ts`). The labels Cha ₪20, Rickshaw ₪60 and Lunch ₪150 are the quick-logs. Widths are in dp, before the row shares out its spare width in proportion.

Width × height (dp)	Size	2-3 quick-logs saved	No quick-logs	After a quick-log (10 min)
250 × 130 (minimum)	compact	+ - Cha ₪20	+ - ▶ 🔍	Undo Keep + -
250 × 180	normal	+ - Cha ₪20	+ - ▶ 🔍	Undo Keep + -
300 × 180	normal	+ Task · - Expense · Cha ₪20	+ - ▶ 🔍	Undo Keep + -
320 × 200	normal	+ Task · - Expense · Cha ₪20	+ - ▶ 🔍	Undo Keep + Task · - Expense
340 × 200 (test's "default")	normal	+ - Cha ₪20 · Rickshaw ₪60	+ - ▶ 🔍	Undo Keep + Task · - Expense
360 × 200	normal	+ - Cha ₪20 · Rickshaw ₪60	+ - ▶ 🔍	Undo Keep + Task · - Expense
400 × 200	normal	+ Task · - Expense · Cha · Rickshaw	+ Task · - Expense · ▶ Focus · 🔍 Tiki	Undo Keep + Task · - Expense
340 × 260	tall	row: + - ▶ 🔍; quick row: all three	+ - ▶ 🔍	Undo row; + - ▶ 🔍

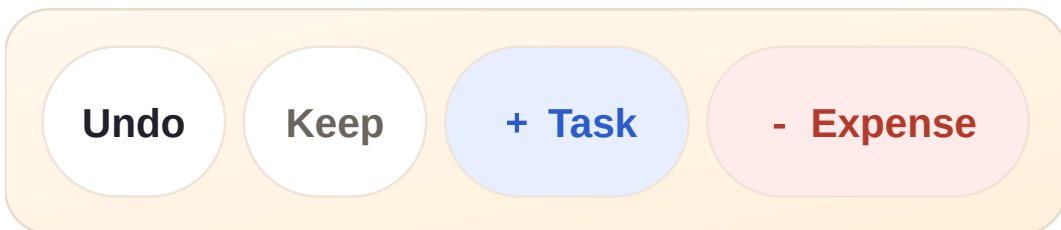
340 × 200 dp, two quick-logs saved (the default 4×2)



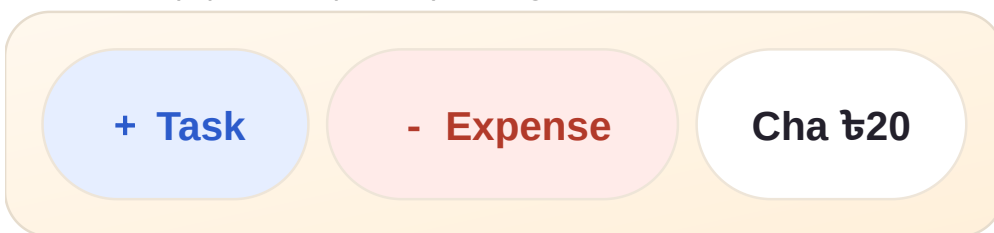
340 × 200 dp, no quick-logs saved yet



340 × 200 dp, for 10 minutes after a quick-log



320 × 200 dp (narrower), two quick-logs saved



Schematic of the widget's bottom row, drawn from the `widgetLayout()` output above with the light widget colours. It is not a screenshot: a launcher may draw it differently. Note that the narrower 320 dp widget shows the names, the default 340 dp one doesn't, and the same pills gain their names for ten minutes after a quick-log (UX-049).

7.3 Mobile ergonomics

- Targets: the scan finds no pressable with a size class under 48 dp that lacks slop. `IconButton` is 48 × 48 (`components/ui/icon-button.tsx:40`) and now carries the month, week, routine and clear buttons. Chips are 40 dp + 4 slop. Small buttons are 44 dp + 4 slop.
- Calendar: 49.1 dp per day on a 360 dp phone, tested.
- Switch rows: only the switch toggles. Tapping the label does nothing (UX-053).
- Sheets: actions stay pinned in the footer. The bill sheet's footer holds two rows of buttons, which is the busiest footer in the app.

7.4 Accessibility

- Contrast: 146 text pairs, 0 below 4.5:1. Lowest: 4.92 (dark: skyText on sky/15 over card). Non-text pairs below 3:1 where 3:1 applies: 5 (listed in Appendix A).
- Roles: 81 `accessibilityRole` and 87 `accessibilityLabel` props. Every `Pressable` has an explicit role or uses `choiceA11y`. 13 `radiogroup` and 2 `tablist` declarations.

- Font scaling: 12 `maxFontSizeMultiplier` caps (chrome only), 4 `adjustsFontSizeToFit`, 0 disabled. The widget's label-width estimate assumes $1 \times \text{font}$ (`widget-layout.ts:56-64`). At large system font sizes labels may truncate with "...", but they no longer clip **Unverified**.
- Motion: 6 files use `useReducedMotion` (unchanged).
- Snackbar: unchanged and good. It follows "Time to take action", is held while TalkBack is on, and announces "Undo available".

7.5 Content and microcopy

- Good: "Your latest changes couldn't be saved on this phone. Keep the app open until they sync." (`lib/outbox.ts:47-49`) is honest and calm. "1 change couldn't be saved on the server" uses the plural helper.
- Wrong in one rare case: a 413 says "Saved on this phone..." after the write was dropped (UX-054).
- Awkward: Discarded "Couldn't log Rent", and the fallback names "Edit in time entries" and "Add in transactions" (UX-054).
- Missing: there is no success line after an export (UX-052).

8. Findings

Open findings at `801dd5f`. New ones are numbered from UX-049. UX-029 keeps its ID. Paths are relative to `apps/mobile/src/` unless they start at the repo root. Within each group, findings are sorted by severity.

8.1 Data trust and recovery

UX-052 · The export offered before deleting an account is deleted when the share sheet returns, and nothing confirms it was saved

Medium User control / feedback · New (side effect of `d1a610c`) · code path verified, share-target behaviour

Unverified · Effort S-M

Where

`features/settings/use-export-data.ts:90-112`; `features/settings/delete-account.tsx:42-55`; `lib/export-file.ts`; `lib/user-data.ts:18-25, 40` (sweep on sign-out); `node_modules/expo-sharing/android/.../SharingModule.kt:52, 80-82` (resolves in `onActivityResult`)

What's wrong

`exportData` writes the full JSON to the cache folder, awaits `Sharing.shareAsync`, and deletes the file in `finally`. On Android, `expo-sharing` resolves when the chooser's activity result comes back. That isn't proof the receiving app has finished reading the `content:// URI`. Gmail sends in the background after the compose screen closes, and Drive queues an upload after "Save". If they read the file after it is gone, the save fails inside the other app, where Tick & Taka can't see it. On top of that, a successful export shows nothing at all, and step "1. Keep a copy" on the Delete-account page has no done state. The user moves on to typing DELETE with no sign that a copy exists.

Evidence

```
const data = await send("GET", "/export");
file = new File(Paths.cache, exportFileName(new Date()));
file.write(JSON.stringify(data));
await Sharing.shareAsync(file.uri, { mimeType: "application/json", ...
});
} catch (error) { notify(friendlyError(error)); }
finally {
  // The share sheet has its copy; ...
  if (file?.exists) file.delete(); // right after the chooser
returns
  setBusy(false); // no success message anywhere
}
```

Why it matters

The export is the only copy of the data outside the server. The Delete-account page offers it as the step before the one irreversible action. A copy that silently failed, followed by a confirmed deletion, is permanent data loss. The privacy reason for deleting the file is sound, but a sweep on sign-out (and so on account deletion) already exists.

Recommendation

1. Don't delete in `finally`. Sweep export files older than an hour when the app starts, and keep the sign-out sweep.
2. Better still, offer "Save to..." through the Storage Access Framework (`expo-file-system`'s directory picker), so the app knows when the write has finished. Then say "Saved tick-taka-export-2026-10-07.json".
3. Show a done state under step 1 ("Copy made · 14:02"). If no copy was made in this visit, word the confirm button's caption accordingly.
4. Check on a device with Drive, Gmail and Files.

UX-054 • Failure copy can contradict what happened

Low Error recovery / microcopy · New (side effect of 801dd5f) · Effort S

Where	<code>lib/error-copy.ts:62</code> ; <code>lib/outbox.ts:55-56</code> ; <code>lib/outbox-failure.ts:5-13</code> ; <code>lib/api.ts:34-37</code> ; <code>components/ui/sync-status.tsx:19-26</code> , 75
What's wrong	(1) A 413, 414 or 431 from Nginx has no API envelope, so it becomes <code>ApiError(413, "http_error")</code> . Since 801dd5f the outbox treats it as final and drops the write. <code>onRejected</code> then words it with <code>friendlyError(error, "save")</code> , which maps every <code>http_error</code> to the offline line: "Couldn't save the task. Saved on this phone. It'll sync when you're back online." The second sentence is false. This is rare: the proxy limit is 10 MB (<code>deploy/nginx-tick-taka.conf:36</code>). (2) Stuck writes are named by their failure label, so Discard reads Discarded "Couldn't log Rent". Unlabelled writes fall back to "Edit in time entries" or "Add in transactions".
Evidence	<pre>// error-copy.ts:62 if (name === "ServerUnreachableError" code === "http_error") return OFFLINE[context]; // outbox-failure.ts:13 (new): T00_LARGE → { status } → classified "reject" → dropped</pre>
Why it matters	The app tells the user a change is safe when it has been thrown away. They will not re-enter it.
Recommendation	In <code>friendlyError</code> , check 413/414/431 before <code>http_error</code> : "That change is too big to send, so it wasn't saved." Give <code>OutboxRequest</code> a noun (what: "Rent payment") apart from its failure <code>label</code> , so the copy reads "Discarded the Rent payment · Undo".

8.2 Widget

UX-049 • At the default size the widget's actions are bare glyphs, even where their names fit

Low Recognition / consistency · New (side effect of the UX-043 fix) · layout computed from the shipped function, rendering `Unverified` · Effort S

Where	<code>features/widget/widget-layout.ts:27</code> (<code>PILL_PADDING = 12</code>), :66-69 (<code>pillWidth</code>), :85-93, :148-166; <code>features/widget/safe-to-spend-widget.tsx:72</code> (row pills get <code>flex</code> and no padding); <code>test/widget-layout.test.ts:37</code> asserts glyphs at the default size
What's wrong	The fit check gives every pill its label width plus 24 dp of padding. In the row, though, pills are laid out with <code>flex</code> and no padding, and the spare width is shared out among them. At 340 × 200 with no quick-logs, the four names need about 223 dp of text plus 18 dp of gaps (241 of 316 dp). The check asks for 337, so a new user sees "+ - ► 🔔". With quick-logs saved, Task and Expense are "+ +" and "- " at 250, 340 and 360 dp, but have their names at 300-320 and 400 dp. So a wider widget can show less text. The same pills also gain their names for ten minutes after a quick-log, then lose them again (section 7.2). In a money widget, "+ +" beside "- " reads as income and expense, but "+ +" opens a new task.

Evidence

```
export function pillWidth(label: string): number {
  return Math.max(TARGET, labelWidth(label) + 2 * PILL_PADDING); // +24
  dp
}
// safe-to-spend-widget.tsx:72
...(flex ? { flex } : { paddingHorizontal: 12 }), // row pills: flex,
no padding
// layouts.ts output, 340×200, no quick-logs:
// task(glyph):48 expense(glyph):48 focus(glyph):48 tiki(glyph):48
```

Why it matters

The widget is the most-seen surface after Today, and principle 2 relies on it. Glyphs make the user remember what each pill does (recognition over recall), and labels that change with size or state look unstable. TalkBack is unaffected, because every pill keeps its full name.

Recommendation

Measure with the padding the pill actually gets: about 8 dp a side, applied as real padding inside the flex pill. Before falling back to glyphs, drop the second quick-log. If glyphs are still needed, don't pair " + " with " - " (for example "✓ Task" → "✓", "Expense" → "✗"). Add tests: "no quick-logs at 340 dp shows names" and "labels never disappear as the widget gets wider".

UX-055 · The widget offers Tiki even when the server's AI is off

Low Dead end · Pre-existing, not raised before · Effort S

Where

features/widget/widget-layout.ts:125, 166 (tiki always in all);
features/widget/widget-cache.ts (no AI flag); in-app rule:
features/assistant/assistant-button.tsx:9-12; reply: features/assistant/use-
assistant.ts:30-32, lib/error-copy.ts:38-39

What's wrong

In the app, Tiki's button only appears when the AI is configured and the assistant feature is on. The widget always includes "👉 Tiki". It shows when no quick-logs are saved and on tall widgets, and it opens the chat already listening. After the user speaks, the answer is "AI isn't set up yet. Use the form instead." or "That AI feature is switched off in Settings > AI."

Recommendation

Write assistant: boolean into the widget cache from useAssistantAvailable(). When it is false, give the slot to Focus or a quick-log.

8.3 Colour meaning and accessibility

UX-050 · Mint is used for controls and for money-out actions, outside its documented role

Low Visual design / principle 4 · Pre-existing (missed in round 3) · Effort S

Where	<code>components/ui/button.tsx:17</code> (<code>money: bg-mint</code>), used by <code>money/recurring-sheet.tsx:153-158</code> ("Paid" on a bill), <code>money/shopping-screen.tsx:223</code> ("Log as one expense"), <code>money/debts-screen.tsx:178-179</code> ("Log repayment"), <code>money/goals-screen.tsx:162-163</code> ("Move to jar"). Hand-rolled switches: <code>money/goal-sheet.tsx:135-140</code> , <code>money/debt-sheet.tsx:182-187</code> , <code>money/budget-edit-sheet.tsx:96-105</code> . Rules: <code>DESIGN.md:24, 34, 74, 76</code> . Test scope: <code>test/colour-meaning.test.ts:66-70</code> .
What's wrong	<code>DESIGN.md</code> gives mint one meaning, "Money in, under budget". It keeps semantic colours to small marks rather than large fills, lists only primary, secondary, time and ghost buttons, and says switches are on = ink. The <code>money</code> variant is a full mint fill used to pay a bill and to log an expense, so money going out is shown in money-in green. The three switches bypass <code>ToggleRow</code> : their "on" track is mint at 2.21:1 on a card in light (2.09 on the background), below the 3:1 <code>DESIGN.md</code> asks for. The thumb position still shows the state. The colour test only covers grape and mango, so none of this is caught.
Why it matters	Principle 4 is a stated product principle. Two switch styles in one app, and green on an expense button, weaken the colour code the money screens depend on.
Recommendation	Remove <code>money</code> . Use <code>primary</code> for the one main action in a sheet, or <code>secondary</code> . Replace the three switches with <code>ToggleRow</code> . Extend <code>colour-meaning.test.ts</code> to mint and coral (an allow-list of money and budget files) and to <code>trackColor</code> . If a money button is wanted, record it in <code>DESIGN.md</code> with its reason.

UX-051 · Plan calendar: task and bill dots are the only visual cue and sit below 3:1 in light

Low Accessibility (WCAG 1.4.11) · Pre-existing (missed in round 3) · Effort S

Where	<code>features/plan/calendar-view.tsx:119, 130-145</code> ; rule <code>DESIGN.md:42</code> ; <code>theme/palette.ts:80-86</code> (<code>SOLE_CUE_MARKS</code>)
What's wrong	A sky dot marks tasks, a 1 px sky ring timed tasks, and a coral dot bills. All three use the bright tokens. In light mode: coral on a card is 2.55:1 and 2.19:1 on today's <code>sky/15</code> cell, and sky on today's cell is 2.72:1 (sky on a card passes at 3.16). <code>DESIGN.md:42</code> says marks that are the only cue use the <code>-text</code> tier. The money calendar follows that (<code>bg-coral-text</code>), but this grid was missed. TalkBack users get the information from the cell's label.
Recommendation	<code>bg-sky-text</code> , <code>border-sky-text</code> , <code>bg-coral-text</code> , as <code>money-calendar.tsx</code> does. Add a test that bright <code>bg-(sky coral mint)</code> marks don't appear in calendar cells, or list this file under the sole-cue rule.

UX-053 · Switch rows only toggle on the switch, and TalkBack meets the label twice

Low Ergonomics / screen reader · New (ToggleRow now carries 15 settings) · **Unverified** · Effort S

Where `components/ui/toggle-row.tsx:18-37`

What's wrong The row is a plain `View`: the label is a `Text` and the `Switch` carries `accessibilityLabel={label}`. Tapping "Billable" does nothing, and only the switch itself, at its native size, toggles. TalkBack will probably stop on "Billable" and then on "Billable, switch, off". On Android, the convention (system Settings, Material lists) is that the whole row toggles.

Recommendation Make the row a `Pressable` with `accessibilityRole="switch"` and `accessibilityState={{ checked: value }}` that toggles on press. Hide the inner `Switch` from accessibility (`importantForAccessibility="no-hide-descendants"`). Use `min-h-14`.

8.4 Content, settings and forms

UX-056 · The time zone the whole app now follows can't be seen or changed in the app

Low Visibility / match with the real world · New (made visible by `1fc2209`) · Effort S-M

Where `lib/use-today.ts:11-19`; `lib/user-time.ts:13-19`;
`packages/shared/src/schemas/settings.ts:25` (default zone); no `timeZone` anywhere in `features/settings/`

What's wrong `1fc2209` correctly made "today" and "this month" follow the user's stored zone on every screen, and `time-zone-use.test.ts` guards it. But nothing shows or sets that zone. It defaults to Asia/Dhaka and never follows the phone. For the stated user, at home in Bangladesh, that is right. On a trip, the Today list, reminders and the budget day stay on Dhaka time with no hint why.

Recommendation Add a "Time zone" field to "Today and planning" that shows the current zone. When the phone's zone differs, add a one-line notice ("Your phone is on Europe/Berlin · Switch"). Record the default in `DESIGN.md` or `PRODUCT.md`.

UX-057 · Three small form dead-ends

Low Error prevention / control · Mixed (new and pre-existing) · Effort S

Where `features/money/goal-sheet.tsx:112-122`; `features/time/time-entry-sheet.tsx:127-133`; `features/habits/habit-sheet.tsx:118-125` with `packages/shared/src/schemas/time.ts:231`

What's wrong (1) Picking today or a past date as a goal's deadline silently keeps the old one (`picked > today ? picked : deadline`), with no message. (2) The time-entry "Area" field shows "No area" as a placeholder but has no `noneLabel`, and `onChange` ignores `null`, so once an area is set it can't be cleared. The task sheet's Area can be cleared. (3) The habit "How often" picker has no `describe`, so a 1× or 7× a week set by Tiki (the schema allows 1-7) shows as a bare "7".

Recommendation (1) Set a minimum date in the picker, or `notify("Pick a day after today")`. (2) `noneLabel="No area"`, passing `null` through. (3) `describe={(n) => `${n}× a week`}`.

8.5 Documentation

UX-029 (carried) · README screenshots predate every fix and aren't dated

Low

Documentation · Not fixed in rounds 2, 3 or 4 · Effort S

Where

docs/screenshots/*.png (23 files, last commit 1f25750, 2026-10-05); README.md:10-14, 33-72

What's wrong

The README still shows the old widget (2.84:1 "Expense" pill, quick-log chips on their own row), the chip-wall task sheet, the 43 dp calendar and the old Today order. Re-shooting needs a device with Play services. One line saying when they were taken would be enough until then.

Recommendation

Under "## Screenshots": "Taken 2026-10-05, before the accessibility, colour and widget changes." Re-shoot once UX-049 is settled.

9. Top 10 quick wins

#	Change	Fixes	Effort
1	Stop deleting the export in <code>finally</code> ; sweep old export files at start-up; show "Copy made · time" under step 1.	UX-052	S
2	Fit widget labels with the padding the pills actually get; drop the second quick-log before glyphing; add the two tests.	UX-049	S
3	Replace the three mint <code>Switches</code> with <code>ToggleRow</code> ; drop the <code>money</code> button variant.	UX-050	S
4	Extend <code>colour-meaning.test.ts</code> to mint and coral and to <code>trackColor</code> .	UX-050	S
5	Use the <code>-text</code> tier for Plan calendar dots and ring.	UX-051	S
6	Word 413/414/431 as "wasn't saved"; give queued writes a noun for the stuck card.	UX-054	S
7	Make the whole <code>ToggleRow</code> a switch.	UX-053	S
8	Hide the widget's Tiki pill when the assistant isn't available.	UX-055	S
9	Goal deadline message, clearable time-entry area, "× a week" label.	UX-057	S
10	One dated line under README › Screenshots.	UX-029	S

10. Three bigger redesign suggestions

R1. Labels first on the widget, measured, not estimated (UX-049, UX-055)

Size classes are now a tested function, which is the right base. The next step is a priority order inside each size: names for Task and Expense first, then one quick-log, then the second quick-log, then glyphs as a last resort. The widths should come from real padding, not an allowance. Two further inputs belong in the cache: whether the assistant is available, and the system font scale (from `PixelRatio.getFontScale()` when the app writes the cache), so that labels are measured at the size they will be drawn. One test per size class should assert what a sighted user reads.

R2. Colour roles for controls, enforced like grape and mango (UX-050, UX-051)

The colour test proved itself on grape and mango. Give each semantic colour an allow-list (mint and coral only in money and budget code, sky for time) and a second rule: bright tokens may not be the only cue in a list or grid cell. Add one "control colours" line to DESIGN.md (buttons: primary, secondary, time, ghost; switches: ink), and let `ToggleRow` and `Button` be the only way to draw them. That turns the remaining residue into compile or test failures.

R3. Make the trust moments explicit (UX-052, UX-054, UX-056)

Three places tell the user something about their data without proof: the export ("did it save?"), a refused write ("saved on this phone"), and the time zone ("whose today is this?"). Treat each as a small status the app owns. The export records when a copy was made and shows it on the Data and Delete pages. Failure copy comes from what the outbox actually did (kept, set aside or dropped), not from the error's shape. The time zone is shown where planning settings live and flagged when the phone disagrees.

11. Strengths (keep these)

- **Fixes that come with tests.** Widget layout (9 tests), calendar width, type scale, colour meaning (now catches classes, palette values and ternaries, and tests its own regex), picker options, pay Undo across a restart, and time-zone use across every screen. Each turns a finding into a rule that fails CI.
- **Types that stop misuse.** `ChipProps` makes `choice` required and forbids `selected` on an action chip, so the round-2 and round-3 role problem can't come back quietly.
- **A complete field vocabulary.** `PickerField` (list or emoji grid, off-list values, none option), `DateField` (with clear), `ToggleRow`, `IconButton` and `MonthStepper` replace about 20 hand-made rows. The sheets now look like one product.
- **Durable Undo.** Pay, receive and skip go through the saved queue, with cancel-before-send or a queued reversal. Discard keeps its group, so Undo restores it exactly.
- **An honest sync story.** The offline, stalled, not-saved-yet, unreadable and stuck states each have their own calm wording, and later edits to a stuck record wait with it.
- **A DESIGN.md that is a real yardstick.** It agrees with itself and with the tokens (0 mismatches), and it records trade-offs with their reasons (FAB, widget sizes, digits, grid bleed).
- **Contrast is solid.** All 146 text pairs pass AA in light, dark, both accent themes and the widget. The lowest is 4.92 (dark: skyText on sky/15 over card).

12. Action plan

Order	ID	Title	Severity	Effort
1	UX-052	Keep the export until the target has it; confirm the copy	Medium	S-M
2	UX-054	True copy for dropped writes; nouns for stuck writes	Low	S
3	UX-049	Widget labels fitted with real padding; stable across sizes	Low	S
4	UX-050	Mint off controls; ToggleRow everywhere; test mint/coral	Low	S
5	UX-051	Plan calendar dots on the -text tier	Low	S
6	UX-053	Whole ToggleRow is the switch	Low	S
7	UX-055	Widget Tiki only when the assistant is available	Low	S
8	UX-057	Three small form dead-ends	Low	S
9	UX-056	Show (and allow changing) the time zone	Low	S-M
10	UX-029	Date or re-shoot the README screenshots	Low	S

10 open findings, ordered by impact over effort. UX-052 comes first because it is the only one that can end in lost data. UX-054 is second because it is the only place the app says something untrue about a change.

Appendix A. WCAG contrast ratios (computed)

Method. WCAG 2.x relative luminance, $(L1 + 0.05) / (L2 + 0.05)$. `contrast.py` parsed `theme/palette.ts` (base and accent themes) and checked it against `global.css` and the DESIGN.md table (0 mismatches in 70 comparisons). It rebuilt the widget palette with the same `blend()` as `widget-colors.ts`, and flattened translucent fills onto the surface below. Thresholds: 4.5 for text, 3.0 for sole-cue marks, control boundaries and switch tracks. "Informative" rows are bright marks that sit beside a label, which DESIGN.md allows, so a ratio under 3 there is not a failure. "n/a" marks decorative hairlines and fills where 1.4.11 doesn't apply. New in round 4: switch tracks and thumbs, the raw mint switches, the Plan calendar's dots, and keypad operators on `line`.

Summary. 254 pairs. Text: 146 pairs, 0 below 4.5:1, lowest 4.92 (dark: skyText on sky/15 over card). Non-text pairs below 3:1 where 3:1 applies (5):

- light: mint (raw Switch on track) on card = **2.21** (switch on-track in goal/debt/budget sheets)
- light: mint (raw Switch on track) on background = **2.09** (budget sheet rows on background)
- light: coral dot on card = **2.55** (plan calendar: bills dot (sole visual cue))
- light: sky dot on today cell (sky/15) = **2.72** (plan calendar: tasks dot on today)
- light: coral dot on today cell (sky/15) = **2.19** (plan calendar: bills dot on today)

All are light-mode bright tokens. They are covered by UX-050 (mint tracks) and UX-051 (calendar dots). Every dark-mode pair passes.

light

Foreground	Background	Hex	Ratio	Needs	Result	Use
ink	background	#23202B / #FFF8EE	15.17	4.5	PASS	text
muted	background	#6B655F / #FFF8EE	5.45	4.5	PASS	text
mangoText	background	#8A5A00 / #FFF8EE	5.62	4.5	PASS	text
skyText	background	#2C5BCB / #FFF8EE	5.75	4.5	PASS	text
mintText	background	#11725D / #FFF8EE	5.55	4.5	PASS	text
coralText	background	#B23A2A / #FFF8EE	5.64	4.5	PASS	text
grapeText	background	#6E44D6 / #FFF8EE	5.75	4.5	PASS	text
ink	card	#23202B / #FFFFFF	15.99	4.5	PASS	text
muted	card	#6B655F / #FFFFFF	5.75	4.5	PASS	text
mangoText	card	#8A5A00 / #FFFFFF	5.93	4.5	PASS	text
skyText	card	#2C5BCB / #FFFFFF	6.06	4.5	PASS	text
mintText	card	#11725D / #FFFFFF	5.85	4.5	PASS	text
coralText	card	#B23A2A / #FFFFFF	5.95	4.5	PASS	text
grapeText	card	#6E44D6 / #FFFFFF	6.07	4.5	PASS	text
skyText	background	#2C5BCB / #FFF8EE	5.75	3.0	PASS	sole-cue mark / control boundary
mintText	background	#11725D / #FFF8EE	5.55	3.0	PASS	sole-cue mark / control boundary
coralText	background	#B23A2A / #FFF8EE	5.64	3.0	PASS	sole-cue mark / control boundary
grapeText	background	#6E44D6 / #FFF8EE	5.75	3.0	PASS	sole-cue mark / control boundary
lineStrong	background	#8F8475 / #FFF8EE	3.48	3.0	PASS	sole-cue mark / control boundary

Foreground	Background	Hex	Ratio	Needs	Result	Use
skyText	card	#2C5BCB / #FFFFFF	6.06	3.0	PASS	sole-cue mark / control boundary
mintText	card	#11725D / #FFFFFF	5.85	3.0	PASS	sole-cue mark / control boundary
coralText	card	#B23A2A / #FFFFFF	5.95	3.0	PASS	sole-cue mark / control boundary
grapeText	card	#6E44D6 / #FFFFFF	6.07	3.0	PASS	sole-cue mark / control boundary
lineStrong	card	#8F8475 / #FFFFFF	3.67	3.0	PASS	sole-cue mark / control boundary
skyText	sky/15 over background	#2C5BCB / #E6E8F1	4.96	4.5	PASS	tinted chip / icon circle
skyText	sky/15 over card	#2C5BCB / #E6EEFF	5.20	4.5	PASS	tinted chip / icon circle
mintText	mint/15 over background	#11725D / #E0F0E2	4.94	4.5	PASS	tinted chip / icon circle
mintText	mint/15 over card	#11725D / #E0F6F1	5.19	4.5	PASS	tinted chip / icon circle
coralText	coral/15 over background	#B23A2A / #FFE5DA	4.95	4.5	PASS	tinted chip / icon circle
coralText	coral/15 over card	#B23A2A / #FFEBE9	5.19	4.5	PASS	tinted chip / icon circle
grapeText	grape/15 over background	#6E44D6 / #F2E5F1	4.99	4.5	PASS	tinted chip / icon circle
grapeText	grape/15 over card	#6E44D6 / #F2EBFF	5.23	4.5	PASS	tinted chip / icon circle
muted	line/60 over background	#6B655F / #F5EDE1	4.95	4.5	PASS	segmented (unselected)
on-mango	mango fill	#23202B / #FFB547	9.10	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	sky fill	#23202B / #5B8CFF	5.06	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	mint fill	#23202B / #2EC4A0	7.25	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	coral fill	#23202B / #FF7A6B	6.28	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	grape fill	#23202B / #A57BFF	5.24	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
background	ink	#FFF8EE / #23202B	15.17	4.5	PASS	selected chip, snackbar text
mangoInverse	ink	#FFB547 / #23202B	9.10	4.5	PASS	snackbar Undo
ink	line (keypad operator, tab pill)	#23202B / #EEE5D8	12.82	4.5	PASS	keypad operator key (now bg-line)
ink	sky/15 over card	#23202B / #E6EEFF	13.74	4.5	PASS	calendar today cell
ink	coral/10 over card	#23202B / #FFF2F0	14.63	4.5	PASS	login error box
muted	card	#6B655F / #FFFFFF	5.75	4.5	PASS	tab label 12px; picker/date placeholder
ink (ToggleRow on track)	card	#23202B / #FFFFFF	15.99	3.0	PASS	switch on-track (ToggleRow)
background (thumb)	ink track	#FFF8EE / #23202B	15.17	3.0	PASS	switch thumb on ink track
lineStrong (off track)	card	#8F8475 / #FFFFFF	3.67	3.0	PASS	switch off-track
mint (raw Switch on track)	card	#2EC4A0 / #FFFFFF	2.21	3.0	FAIL	switch on-track in goal/debt/budget sheets

Foreground	Background	Hex	Ratio	Needs	Result	Use
mint (raw Switch on track)	background	#2EC4A0 / #FFF8EE	2.09	3.0	FAIL	budget sheet rows on background
sky dot	card	#5B8CFF / #FFFFFF	3.16	3.0	PASS	plan calendar: tasks dot (sole visual cue)
coral dot	card	#FF7A6B / #FFFFFF	2.55	3.0	FAIL	plan calendar: bills dot (sole visual cue)
sky dot	today cell (sky/15)	#5B8CFF / #E6EEFF	2.72	3.0	FAIL	plan calendar: tasks dot on today
coral dot	today cell (sky/15)	#FF7A6B / #E6EEFF	2.19	3.0	FAIL	plan calendar: bills dot on today
coralText bar	card	#B23A2A / #FFFFFF	5.95	3.0	PASS	money calendar spend bar
mango (bright)	background	#FFB547 / #FFF8EE	1.67	3.0*	below 3 (informative)	bright mark beside a label (informative)
mango (bright)	card	#FFB547 / #FFFFFF	1.76	3.0*	below 3 (informative)	bright mark beside a label (informative)
sky (bright)	background	#5B8CFF / #FFF8EE	3.00	3.0*	below 3 (informative)	bright mark beside a label (informative)
sky (bright)	card	#5B8CFF / #FFFFFF	3.16	3.0*	PASS	bright mark beside a label (informative)
mint (bright)	background	#2EC4A0 / #FFF8EE	2.09	3.0*	below 3 (informative)	bright mark beside a label (informative)
mint (bright)	card	#2EC4A0 / #FFFFFF	2.21	3.0*	below 3 (informative)	bright mark beside a label (informative)
coral (bright)	background	#FF7A6B / #FFF8EE	2.41	3.0*	below 3 (informative)	bright mark beside a label (informative)
coral (bright)	card	#FF7A6B / #FFFFFF	2.55	3.0*	below 3 (informative)	bright mark beside a label (informative)
grape (bright)	background	#A57BFF / #FFF8EE	2.90	3.0*	below 3 (informative)	bright mark beside a label (informative)
grape (bright)	card	#A57BFF / #FFFFFF	3.05	3.0*	PASS	bright mark beside a label (informative)
line	background	#EEE5D8 / #FFF8EE	1.18	—	n/a	hairline (decorative)
line	card	#EEE5D8 / #FFFFFF	1.25	—	n/a	hairline (decorative)

light / mint-breeze

Foreground	Background	Hex	Ratio	Needs	Result	Use
ink	background	#23202B / #ECFAF5	14.90	4.5	PASS	text
muted	background	#6B655F / #ECFAF5	5.36	4.5	PASS	text
mangoText	background	#8A5A00 / #ECFAF5	5.52	4.5	PASS	text
skyText	background	#2C5BCB / #ECFAF5	5.64	4.5	PASS	text
mintText	background	#11725D / #ECFAF5	5.45	4.5	PASS	text
coralText	background	#B23A2A / #ECFAF5	5.54	4.5	PASS	text
grapeText	background	#6E44D6 / #ECFAF5	5.65	4.5	PASS	text
ink	card	#23202B / #FFFFFF	15.99	4.5	PASS	text
muted	card	#6B655F / #FFFFFF	5.75	4.5	PASS	text
mangoText	card	#8A5A00 / #FFFFFF	5.93	4.5	PASS	text
skyText	card	#2C5BCB / #FFFFFF	6.06	4.5	PASS	text
mintText	card	#11725D / #FFFFFF	5.85	4.5	PASS	text
coralText	card	#B23A2A / #FFFFFF	5.95	4.5	PASS	text

Foreground	Background	Hex	Ratio	Needs	Result	Use
grapeText	card	#6E44D6 / #FFFFFF	6.07	4.5	PASS	text
skyText	background	#2C5BCB / #ECFAF5	5.64	3.0	PASS	sole-cue mark / control boundary
mintText	background	#11725D / #ECFAF5	5.45	3.0	PASS	sole-cue mark / control boundary
coralText	background	#B23A2A / #ECFAF5	5.54	3.0	PASS	sole-cue mark / control boundary
grapeText	background	#6E44D6 / #ECFAF5	5.65	3.0	PASS	sole-cue mark / control boundary
lineStrong	background	#8F8475 / #ECFAF5	3.42	3.0	PASS	sole-cue mark / control boundary
skyText	card	#2C5BCB / #FFFFFF	6.06	3.0	PASS	sole-cue mark / control boundary
mintText	card	#11725D / #FFFFFF	5.85	3.0	PASS	sole-cue mark / control boundary
coralText	card	#B23A2A / #FFFFFF	5.95	3.0	PASS	sole-cue mark / control boundary
grapeText	card	#6E44D6 / #FFFFFF	6.07	3.0	PASS	sole-cue mark / control boundary
lineStrong	card	#8F8475 / #FFFFFF	3.67	3.0	PASS	sole-cue mark / control boundary

light / grape-dusk

Foreground	Background	Hex	Ratio	Needs	Result	Use
ink	background	#23202B / #F6F1FF	14.43	4.5	PASS	text
muted	background	#6B655F / #F6F1FF	5.19	4.5	PASS	text
mangoText	background	#8A5A00 / #F6F1FF	5.35	4.5	PASS	text
skyText	background	#2C5BCB / #F6F1FF	5.47	4.5	PASS	text
mintText	background	#11725D / #F6F1FF	5.28	4.5	PASS	text
coralText	background	#B23A2A / #F6F1FF	5.36	4.5	PASS	text
grapeText	background	#6E44D6 / #F6F1FF	5.47	4.5	PASS	text
ink	card	#23202B / #FFFFFF	15.99	4.5	PASS	text
muted	card	#6B655F / #FFFFFF	5.75	4.5	PASS	text
mangoText	card	#8A5A00 / #FFFFFF	5.93	4.5	PASS	text
skyText	card	#2C5BCB / #FFFFFF	6.06	4.5	PASS	text
mintText	card	#11725D / #FFFFFF	5.85	4.5	PASS	text
coralText	card	#B23A2A / #FFFFFF	5.95	4.5	PASS	text
grapeText	card	#6E44D6 / #FFFFFF	6.07	4.5	PASS	text
skyText	background	#2C5BCB / #F6F1FF	5.47	3.0	PASS	sole-cue mark / control boundary
mintText	background	#11725D / #F6F1FF	5.28	3.0	PASS	sole-cue mark / control boundary
coralText	background	#B23A2A / #F6F1FF	5.36	3.0	PASS	sole-cue mark / control boundary
grapeText	background	#6E44D6 / #F6F1FF	5.47	3.0	PASS	sole-cue mark / control boundary
lineStrong	background	#8F8475 / #F6F1FF	3.31	3.0	PASS	sole-cue mark / control boundary
skyText	card	#2C5BCB / #FFFFFF	6.06	3.0	PASS	sole-cue mark / control boundary
mintText	card	#11725D / #FFFFFF	5.85	3.0	PASS	sole-cue mark / control boundary
coralText	card	#B23A2A / #FFFFFF	5.95	3.0	PASS	sole-cue mark / control boundary
grapeText	card	#6E44D6 / #FFFFFF	6.07	3.0	PASS	sole-cue mark / control boundary
lineStrong	card	#8F8475 / #FFFFFF	3.67	3.0	PASS	sole-cue mark / control boundary

dark

Foreground	Background	Hex	Ratio	Needs	Result	Use
ink	background	#F4F1EA / #16151C	16.07	4.5	PASS	text
muted	background	#A09AA6 / #16151C	6.62	4.5	PASS	text
mangoText	background	#FFC266 / #16151C	11.36	4.5	PASS	text
skyText	background	#7AA2FF / #16151C	7.29	4.5	PASS	text
mintText	background	#4FD8B5 / #16151C	10.21	4.5	PASS	text

Foreground	Background	Hex	Ratio	Needs	Result	Use
coralText	background	#FF9385 / #16151C	8.43	4.5	PASS	text
grapeText	background	#B996FF / #16151C	7.66	4.5	PASS	text
ink	card	#F4F1EA / #22202B	14.22	4.5	PASS	text
muted	card	#A09AA6 / #22202B	5.86	4.5	PASS	text
mangoText	card	#FFC266 / #22202B	10.05	4.5	PASS	text
skyText	card	#7AA2FF / #22202B	6.44	4.5	PASS	text
mintText	card	#4FD8B5 / #22202B	9.03	4.5	PASS	text
coralText	card	#FF9385 / #22202B	7.46	4.5	PASS	text
grapeText	card	#B996FF / #22202B	6.77	4.5	PASS	text
skyText	background	#7AA2FF / #16151C	7.29	3.0	PASS	sole-cue mark / control boundary
mintText	background	#4FD8B5 / #16151C	10.21	3.0	PASS	sole-cue mark / control boundary
coralText	background	#FF9385 / #16151C	8.43	3.0	PASS	sole-cue mark / control boundary
grapeText	background	#B996FF / #16151C	7.66	3.0	PASS	sole-cue mark / control boundary
lineStrong	background	#7A748C / #16151C	4.06	3.0	PASS	sole-cue mark / control boundary
skyText	card	#7AA2FF / #22202B	6.44	3.0	PASS	sole-cue mark / control boundary
mintText	card	#4FD8B5 / #22202B	9.03	3.0	PASS	sole-cue mark / control boundary
coralText	card	#FF9385 / #22202B	7.46	3.0	PASS	sole-cue mark / control boundary
grapeText	card	#B996FF / #22202B	6.77	3.0	PASS	sole-cue mark / control boundary
lineStrong	card	#7A748C / #22202B	3.59	3.0	PASS	sole-cue mark / control boundary
skyText	sky/15 over background	#7AA2FF / #252A3E	5.70	4.5	PASS	tinted chip / icon circle
skyText	sky/15 over card	#7AA2FF / #2F344B	4.92	4.5	PASS	tinted chip / icon circle
mintText	mint/15 over background	#4FD8B5 / #1F3233	7.57	4.5	PASS	tinted chip / icon circle
mintText	mint/15 over card	#4FD8B5 / #293C40	6.51	4.5	PASS	tinted chip / icon circle
coralText	coral/15 over background	#FF9385 / #39282C	6.45	4.5	PASS	tinted chip / icon circle
coralText	coral/15 over card	#FF9385 / #433138	5.63	4.5	PASS	tinted chip / icon circle
grapeText	grape/15 over background	#B996FF / #2E283E	5.96	4.5	PASS	tinted chip / icon circle
grapeText	grape/15 over card	#B996FF / #39324B	5.12	4.5	PASS	tinted chip / icon circle
muted	line/60 over background	#A09AA6 / #282631	5.43	4.5	PASS	segmented (unselected)
on-mango	mango fill	#23202B / #FFC266	10.02	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	sky fill	#23202B / #7AA2FF	6.43	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	mint fill	#23202B / #4FD8B5	9.00	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	coral fill	#23202B / #FF9385	7.44	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
on-mango	grape fill	#23202B / #B996FF	6.76	4.5	PASS	text/icon on fill (primary, money button, swipe tray)
background	ink	#16151C / #F4F1EA	16.07	4.5	PASS	selected chip, snackbar text
mangoInverse	ink	#8A5A00 / #F4F1EA	5.25	4.5	PASS	snackbar Undo
ink	line (keypad operator, tab pill)	#F4F1EA / #34313F	11.23	4.5	PASS	keypad operator key (now bg-line)
ink	sky/15 over card	#F4F1EA / #2F344B	10.86	4.5	PASS	calendar today cell
ink	coral/10 over card	#F4F1EA / #382C34	11.80	4.5	PASS	login error box

Foreground	Background	Hex	Ratio	Needs	Result	Use
muted	card	#A09AA6 / #22202B	5.86	4.5	PASS	tab label 12px; picker/date placeholder
ink (ToggleRow on track)	card	#F4F1EA / #22202B	14.22	3.0	PASS	switch on-track (ToggleRow)
background (thumb)	ink track	#16151C / #F4F1EA	16.07	3.0	PASS	switch thumb on ink track
lineStrong (off track)	card	#7A748C / #22202B	3.59	3.0	PASS	switch off-track
mint (raw Switch on track)	card	#4FD8B5 / #22202B	9.03	3.0	PASS	switch on-track in goal/debt/budget sheets
mint (raw Switch on track)	background	#4FD8B5 / #16151C	10.21	3.0	PASS	budget sheet rows on background
sky dot	card	#7AA2FF / #22202B	6.44	3.0	PASS	plan calendar: tasks dot (sole visual cue)
coral dot	card	#FF9385 / #22202B	7.46	3.0	PASS	plan calendar: bills dot (sole visual cue)
sky dot	today cell (sky/15)	#7AA2FF / #2F344B	4.92	3.0	PASS	plan calendar: tasks dot on today
coral dot	today cell (sky/15)	#FF9385 / #2F344B	5.70	3.0	PASS	plan calendar: bills dot on today
coralText bar	card	#FF9385 / #22202B	7.46	3.0	PASS	money calendar spend bar
mango (bright)	background	#FFC266 / #16151C	11.36	3.0*	PASS	bright mark beside a label (informative)
mango (bright)	card	#FFC266 / #22202B	10.05	3.0*	PASS	bright mark beside a label (informative)
sky (bright)	background	#7AA2FF / #16151C	7.29	3.0*	PASS	bright mark beside a label (informative)
sky (bright)	card	#7AA2FF / #22202B	6.44	3.0*	PASS	bright mark beside a label (informative)
mint (bright)	background	#4FD8B5 / #16151C	10.21	3.0*	PASS	bright mark beside a label (informative)
mint (bright)	card	#4FD8B5 / #22202B	9.03	3.0*	PASS	bright mark beside a label (informative)
coral (bright)	background	#FF9385 / #16151C	8.43	3.0*	PASS	bright mark beside a label (informative)
coral (bright)	card	#FF9385 / #22202B	7.46	3.0*	PASS	bright mark beside a label (informative)
grape (bright)	background	#B996FF / #16151C	7.66	3.0*	PASS	bright mark beside a label (informative)
grape (bright)	card	#B996FF / #22202B	6.77	3.0*	PASS	bright mark beside a label (informative)
line	background	#34313F / #16151C	1.43	—	n/a	hairline (decorative)
line	card	#34313F / #22202B	1.27	—	n/a	hairline (decorative)

dark / mint-breeze

Foreground	Background	Hex	Ratio	Needs	Result	Use
ink	background	#F4F1EA / #101C1A	15.47	4.5	PASS	text
muted	background	#A09AA6 / #101C1A	6.37	4.5	PASS	text
mangoText	background	#FFC266 / #101C1A	10.94	4.5	PASS	text
skyText	background	#7AA2FF / #101C1A	7.02	4.5	PASS	text
mintText	background	#4FD8B5 / #101C1A	9.83	4.5	PASS	text
coralText	background	#FF9385 / #101C1A	8.12	4.5	PASS	text

Foreground	Background	Hex	Ratio	Needs	Result	Use
grapeText	background	#B996FF / #101C1A	7.37	4.5	PASS	text
ink	card	#F4F1EA / #1A2825	13.55	4.5	PASS	text
muted	card	#A09AA6 / #1A2825	5.58	4.5	PASS	text
mangoText	card	#FFC266 / #1A2825	9.58	4.5	PASS	text
skyText	card	#7AA2FF / #1A2825	6.14	4.5	PASS	text
mintText	card	#4FD8B5 / #1A2825	8.60	4.5	PASS	text
coralText	card	#FF9385 / #1A2825	7.11	4.5	PASS	text
grapeText	card	#B996FF / #1A2825	6.45	4.5	PASS	text
skyText	background	#7AA2FF / #101C1A	7.02	3.0	PASS	sole-cue mark / control boundary
mintText	background	#4FD8B5 / #101C1A	9.83	3.0	PASS	sole-cue mark / control boundary
coralText	background	#FF9385 / #101C1A	8.12	3.0	PASS	sole-cue mark / control boundary
grapeText	background	#B996FF / #101C1A	7.37	3.0	PASS	sole-cue mark / control boundary
lineStrong	background	#7A748C / #101C1A	3.91	3.0	PASS	sole-cue mark / control boundary
skyText	card	#7AA2FF / #1A2825	6.14	3.0	PASS	sole-cue mark / control boundary
mintText	card	#4FD8B5 / #1A2825	8.60	3.0	PASS	sole-cue mark / control boundary
coralText	card	#FF9385 / #1A2825	7.11	3.0	PASS	sole-cue mark / control boundary
grapeText	card	#B996FF / #1A2825	6.45	3.0	PASS	sole-cue mark / control boundary
lineStrong	card	#7A748C / #1A2825	3.42	3.0	PASS	sole-cue mark / control boundary

dark / grape-dusk

Foreground	Background	Hex	Ratio	Needs	Result	Use
ink	background	#F4F1EA / #181424	15.99	4.5	PASS	text
muted	background	#A09AA6 / #181424	6.59	4.5	PASS	text
mangoText	background	#FFC266 / #181424	11.30	4.5	PASS	text
skyText	background	#7AA2FF / #181424	7.25	4.5	PASS	text
mintText	background	#4FD8B5 / #181424	10.15	4.5	PASS	text
coralText	background	#FF9385 / #181424	8.39	4.5	PASS	text
grapeText	background	#B996FF / #181424	7.62	4.5	PASS	text
ink	card	#F4F1EA / #241E34	14.21	4.5	PASS	text
muted	card	#A09AA6 / #241E34	5.85	4.5	PASS	text
mangoText	card	#FFC266 / #241E34	10.04	4.5	PASS	text
skyText	card	#7AA2FF / #241E34	6.44	4.5	PASS	text
mintText	card	#4FD8B5 / #241E34	9.02	4.5	PASS	text
coralText	card	#FF9385 / #241E34	7.45	4.5	PASS	text
grapeText	card	#B996FF / #241E34	6.77	4.5	PASS	text
skyText	background	#7AA2FF / #181424	7.25	3.0	PASS	sole-cue mark / control boundary
mintText	background	#4FD8B5 / #181424	10.15	3.0	PASS	sole-cue mark / control boundary
coralText	background	#FF9385 / #181424	8.39	3.0	PASS	sole-cue mark / control boundary
grapeText	background	#B996FF / #181424	7.62	3.0	PASS	sole-cue mark / control boundary
lineStrong	background	#7A748C / #181424	4.04	3.0	PASS	sole-cue mark / control boundary
skyText	card	#7AA2FF / #241E34	6.44	3.0	PASS	sole-cue mark / control boundary
mintText	card	#4FD8B5 / #241E34	9.02	3.0	PASS	sole-cue mark / control boundary
coralText	card	#FF9385 / #241E34	7.45	3.0	PASS	sole-cue mark / control boundary
grapeText	card	#B996FF / #241E34	6.77	3.0	PASS	sole-cue mark / control boundary
lineStrong	card	#7A748C / #241E34	3.59	3.0	PASS	sole-cue mark / control boundary

widget light

Foreground	Background	Hex	Ratio	Needs	Result	Use
muted 13	from	#6B655F / #FFF8EE	5.45	4.5	PASS	label, status line
mintText 26 bold	from	#11725D / #FFF8EE	5.55	4.5	PASS	hero
coralText 26 bold	from	#B23A2A / #FFF8EE	5.64	4.5	PASS	hero (over)
muted 13	to	#6B655F / #FFF0DA	5.13	4.5	PASS	label, status line
mintText 26 bold	to	#11725D / #FFF0DA	5.22	4.5	PASS	hero
coralText 26 bold	to	#B23A2A / #FFF0DA	5.30	4.5	PASS	hero (over)
ink	card	#23202B / #FFFFFF	15.99	4.5	PASS	next-up title, quick-log, Undo, Tiki pills
muted	card	#6B655F / #FFFFFF	5.75	4.5	PASS	Next up label, Keep pill
skyText 13 bold	card	#2C5BCB / #FFFFFF	6.06	4.5	PASS	next-up time
skyText 13 bold	skyTint	#2C5BCB / #E6EEFF	5.20	4.5	PASS	Task / Focus pill (label or + glyph)
coralText 13 bold	coralTint	#B23A2A / #FFEBE9	5.19	4.5	PASS	Expense pill (label or - glyph)
mint bar	track	#2EC4A0 / #EEE5D8	1.77	—	n/a	spend bar (beside text)
card pill	to	#FFFFFF / #FFF0DA	1.12	—	n/a	pill edge (label identifies it)

widget dark

Foreground	Background	Hex	Ratio	Needs	Result	Use
muted 13	from	#A09AA6 / #22202B	5.86	4.5	PASS	label, status line
mintText 26 bold	from	#4FD8B5 / #22202B	9.03	4.5	PASS	hero
coralText 26 bold	from	#FF9385 / #22202B	7.46	4.5	PASS	hero (over)
muted 13	to	#A09AA6 / #16151C	6.62	4.5	PASS	label, status line
mintText 26 bold	to	#4FD8B5 / #16151C	10.21	4.5	PASS	hero
coralText 26 bold	to	#FF9385 / #16151C	8.43	4.5	PASS	hero (over)
ink	card	#F4F1EA / #2A2833	12.84	4.5	PASS	next-up title, quick-log, Undo, Tiki pills
muted	card	#A09AA6 / #2A2833	5.29	4.5	PASS	Next up label, Keep pill
skyText 13 bold	card	#7AA2FF / #2A2833	5.82	4.5	PASS	next-up time
skyText 13 bold	skyTint	#7AA2FF / #2F344B	4.92	4.5	PASS	Task / Focus pill (label or + glyph)
coralText 13 bold	coralTint	#FF9385 / #433138	5.63	4.5	PASS	Expense pill (label or - glyph)
mint bar	track	#4FD8B5 / #34313F	7.13	—	n/a	spend bar (beside text)
card pill	to	#2A2833 / #16151C	1.25	—	n/a	pill edge (label identifies it)

* Needs 3.0 only if the mark is the only cue.

Appendix B. Accessibility and touch-target scan

Check (grep / parse over apps/mobile/src)	R3	R4	Note
<Pressable> + <SwipeRowPressable> tags	71	60	45 files. Fewer, mainly because IconButton, PickerField and DateField replaced hand-made pressables (my parser differs slightly from round 3's).
Pressable without an explicit role or choiceA11y	4	0	The round-3 four were accessible={false} rows; none are left.
Tags without an accessibilityLabel attribute	—	26	All have visible text children, which TalkBack reads.
accessibilityRole / accessibilityLabel	84 / 92	81 / 87	Lower count, same coverage: shared components now carry them.
choiceA11y(· radiogroup · tablist	3 · 4 · 1	5 · 13 · 2	Every single-choice row sits in a radiogroup; the tab bar is a tablist.
<Chip> with choice / without	13 / 28	15 / 0	Required by type: 11 single, 1 multi, 3 action.
PickerField · DateField · ToggleRow · IconButton · MonthStepper	—	47 · 9 · 15 · 10 · 3	New shared field components.
Hand-rolled <Switch>	—	4	1 inside ToggleRow; 3 in money sheets with a mint track (UX-050).
hitSlop	14	12	Calendar slop removed (UX-040 fixed).
Pressables with a size class under 48 dp and no slop	—	0	
maxFontSizeMultiplier · adjustsFontSizeToFit · allowFontScaling={false}	12 · 4 · 0	12 · 4 · 0	Chrome only.
Text below 13 px	2	1	tab-bar.tsx:77 (12 px, allowed); chart axes now 13.
useReducedMotion(calls (files)	(6)	8 (6)	Unchanged.
announceForAccessibility · getRecommendedTimeoutMillis · isScreenReaderEnabled	2 · 1 · 1	2 · 1 · 1	Snackbar.
accessibilityLiveRegion · accessibilityHint	7 · 14	7 · 14	
Alert.alert	4	4	Sign-out (2), New chat, account action.

Targets after this round

Location	Size (dp)	Status
Calendar cells (calendar-grid.ts ; both calendars)	49 × 56-64	Fixed, tested
IconButton (month/week arrows, routine steps, clear)	48 × 48	OK

Location	Size (dp)	Status
PickerField / DateField	≥ 56 tall; options ≥ 48	OK
Chips	40 + 4 slop	OK
Small buttons (Retry/Discard, Paid/Received)	44 + 4 slop; Retry–Discard gap 24	OK
Widget pills	48 tall, ≥ 48 wide	OK (labels: UX-049)
ToggleRow	row 48; only the native switch toggles	UX-053 Unverified
Snackbar action	40 + 12 slop	OK

Appendix C. Requirement checklist (UX-relevant)

Requirement (source)	R3	R4	Ref
Log anything in under 5 s (PRODUCT)	Met by design; widget quick-log may need a resize	Met by design; widget quick-log at default size	\$5, UX-043 fixed
Targets ≥ 48 dp (PRODUCT)	Met except calendar width	Met (switch row: native switch only)	App. B, UX-053
WCAG AA in light and dark (PRODUCT)	Met in app, accent themes and widget	Met for all text; 5 light non-text marks under 3:1	App. A
Dark text on mango and every fill	Met	Met	App. A
Colour never the only signal	Met	Mostly: Plan calendar dots	UX-051
Colour means something (PRODUCT, DESIGN)	Met, small residue	Met for grape/mango; mint residue	UX-050
Reduce motion respected	Met	Met (device Unverified)	App. B
Tabular digits for amounts and timers	Met	Met	—
Delete → Undo snackbar, not a dialog (spec)	Met	Met; Discard too	UX-046 fixed
Pickers in sheets instead of chip rows (DESIGN)	Partial (5 sheets)	Met	UX-013 fixed
Caption 13 is the smallest text (DESIGN)	Met except chart axes	Met, tested	UX-047 fixed
Chips: action, radio or checkbox; switches for on/off (DESIGN)	—	Met by type; 3 switches off-style	UX-036 fixed, UX-050
Month grids ≥ 48 dp per day (DESIGN)	—	Met, tested	UX-040 fixed
Widget per size class (DESIGN)	—	Met as written; label fit too strict	UX-049
Guilt-free UI (PRODUCT)	Met	Met	—
Bangla and English; digits scope stated (README, DESIGN)	Met	Met, now in DESIGN.md	DESIGN.md:47
Offline: changes queue, then sync (README)	Met; pay-undo gap	Met; pay Undo durable	UX-042 fixed
Full JSON export (README)	Met	Met; hand-off and confirmation weak	UX-052

Appendix D. Files reviewed and commands

```
git log --oneline 7e2bfeb..HEAD # 36 commits
git diff --stat 7e2bfeb..HEAD -- apps/mobile/src DESIGN.md README.md
git diff 7e2bfeb..HEAD -- DESIGN.md README.md
git show 801dd5f c6a8012 2372a63 (--stat, diff); git blame
apps/mobile/src/components/ui/button.tsx
git show 7e2bfeb:apps/mobile/src/features/time/time-entry-sheet.tsx
git log -- docs/screenshots DESIGN.md apps/mobile/src/app/login.tsx
python3 -I _ux_tmp/contrast.py → 254 pairs, 0 token mismatches / 70, 0 text pairs < 4.5
python3 -I _ux_tmp/scan.py → 60 pressables in 45 files; role/label/size; prop counts;
text < 13
bun run _ux_tmp/layouts.ts → widgetLayout() for 9 sizes × 4 states (imports the shipped
module)
grep: Chip call sites and choice; radiogroup/tablist; <Switch / trackColor; variant="money";
bg-(sky|coral|mint) in calendar cells; timeZone in features/settings; isExportFile;
expo-sharing SharingModule.kt (startActivityForResult / onActivityResult)
Reused from _raw/: lint.txt (Biome 555 files, exit 0), typecheck.txt (exit 0),
test.txt (shared 247, mobile 291, api 236 pass; 0 fail)
Read: PRODUCT.md, DESIGN.md, README.md; theme/palette.ts, global.css, tailwind.config.js;
components/ui/{chip, ally-actions, segmented, picker-field, picker-options, date-field,
toggle-row, icon-button, month-stepper, chart-axis, calendar-grid, card, button, snackbar,
sync-status}; components/navigation/tab-bar.tsx; features/widget/*;
features/today/next-up.tsx; features/money/{recurring-pay, recurring-sheet, account-sheet,
goal-sheet, debt-sheet, budget-edit-sheet, money-calendar, events-screen, goals-screen,
debts-screen, shopping-screen}; features/plan/{calendar-view, week-planner};
features/tasks/{task-sheet, task-more-fields}; features/time/{time-entry-sheet,
time-tracking, focus-stats}; features/review/{insights-screen, ai-usage-screen};
features/settings/{delete-account, delete-account-rules, use-export-data, data-section,
account-section, setting-row, planning-settings, notification-settings};
features/habits/habit-sheet.tsx; features/quick-add/quick-add-sheet.tsx;
features/assistant/{assistant-button, money-drafts}; app/{assistant, (tabs)/index}.tsx;
lib/{outbox, outbox-queue, outbox-failure, error-copy, api, http, notify, snack-timing,
use-today, user-time, local-day, export-file, user-data, receipts};
test/{widget-layout, colour-meaning, type-scale, calendar-grid, picker-options,
recurring-pay, time-zone-use, contrast, outbox-queue}.test.ts; apps/mobile/app.json;
apps/api/src/modules/recurring/{routes, service}.ts; deploy/nginx-tick-taka.conf
```